

Association of Hemispheric Laterality with Neurosurgery in ICH: A Post-Hoc Bayesian Analysis of ATACH-2 and ERICH

Authors: Nikhil M Patel, MD, MBA^{1 2}; Rahul R Karamchandani, MD²; Matthew N Jaffa, DO; Liang Wang, PhD⁴; Hongmei Yang, PhD⁴; Rhonda Cadena, MD¹; Christa Swisher, MD²; Joe D Bernard, MD⁵; Jonathan M Parish, MD⁵; William R Stetler, MD⁵; Rick Magun, MD⁶; Aaron Taylor, MD⁶; Daniel Woo, MD⁶; Adnan Qureshi, MD⁷; Andrew W Asimos, MD⁸; Nicholas A Morris, MD⁹

Affiliations:

¹ Department of Internal Medicine, Wake Forest School of Medicine, Advocate Health, Charlotte, NC, USA.

² Department of Neurology, Neurosciences Institute, Wake Forest School of Medicine, Advocate Health, Charlotte, NC, USA.

³ Division of Neurocritical Care, Department of Neurology, Ayer Neuroscience Institute, Hartford Hospital, Hartford, CT, USA.

⁴ Clinical Quality Analytics, Advocate Health, Charlotte, NC 28203, USA.

⁵ Carolina Neurosurgery & Spine Associates, Department of Neurosurgery, Neurosciences Institute, Wake Forest University School of Medicine, Advocate Health, Charlotte, NC, USA.

⁶ Department of Neurology, University at Buffalo Jacobs School of Medicine & Biomedical Sciences, Buffalo, NY, USA.

⁷ Zeenat Qureshi Stroke Institute and Department of Neurology, University of Missouri-Columbia, Columbia, MO, USA.

⁸ Department of Emergency Medicine, Neurosciences Institute, Advocate Health, Charlotte, NC, USA.

⁹ Department of Neurology, Program in Trauma, University of Maryland School of Medicine, Baltimore, MD, USA.

Corresponding Author:

Nikhil M Patel, MD, MBA

Department of Internal Medicine

Department of Neurology

Wake Forest School of Medicine

1000 Blythe Blvd.

Charlotte, NC 28203

nikhil.m.patel@wfusm.edu

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Author contributions:

Nikhil Patel had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Concept and design: Patel, Jaffa, Karamchandani, Morris

Acquisition, analysis, or interpretation of data: Patel, Wang, Yang, Woo, Qureshi

Drafting of manuscript: Patel, Jaffa, Karamchandani

Critical review of the manuscript for important intellectual content: Patel, Karamchandani, Jaffa, Cadena, Swisher, Bernard, Parish, Stetler, Magun, Taylor, Woo, Qureshi, Asimos, Morris

Statistical analysis: Patel, Wang, Yang

Supervision: Woo, Qureshi, Asimos, Morris

Abstract

Background: Hemispheric laterality influences clinical management in stroke. Evidence in ischemic stroke suggests clinicians may hesitate to offer aggressive interventions for dominant-hemisphere injuries. We evaluated the effect of hemispheric lateralization of intracerebral hemorrhage (ICH) on aggressive surgical intervention and 90-day functional outcomes.

Methods: This post-hoc analysis included 3,477 patients with supratentorial ICH from the ATACH-2 and ERICh studies. The primary outcome was neurosurgical intervention (hematoma evacuation, decompressive craniectomy, or other craniotomy). Secondary outcomes included external ventricular drain (EVD) placement, 90-day modified Rankin Scale (mRS), and EuroQOL domains. Hierarchical Bayesian regression models were adjusted for age, admission Glasgow Coma Scale, ICH location, ICH volume, and intraventricular hemorrhage.

Results: Of 3,477 patients, 1,720 (49%) had left hemispheric ICH (LH-ICH). Right hemispheric ICH (RH-ICH) was associated with higher odds of neurosurgical intervention (8.2% vs. 5.0%; aOR, 1.56 [95% CrI, 1.07–2.26]; posterior probability aOR > 1: 99%) and EVD placement (15.5% vs. 13.7%; aOR, 1.34 [95% CrI, 1.01–1.76]; posterior probability aOR > 1: 98%). Despite receiving less aggressive care, LH-ICH patients demonstrated better functional outcomes; RH-ICH was associated with worse 90-day mRS (aOR, 1.18 [95% CrI, 1.01–1.36]; posterior probability aOR > 1: 98%) and worse EuroQOL mobility (aOR, 1.43 [95% CrI, 1.2–1.71]; posterior probability aOR > 1: 100%) and self-care (aOR, 1.43 [95% CrI, 1.2–1.72]; posterior probability aOR > 1: 100%).

Conclusions: Patients with LH-ICH were less likely to receive aggressive surgical care yet demonstrated superior 90-day functional outcomes. These findings suggest potential bias in ICH treatment decisions based on hemispheric laterality, highlighting the need to ensure equitable access to beneficial interventions.

Trial Registration: Clinicaltrials.gov Identifier: ATACH-2, NCT01176565; ERICH, NCT01202864

Introduction

Intracerebral hemorrhage (ICH) remains a challenging disease, marked by its sudden onset, morbidity, and mortality.^{1,2} While management of ischemic stroke has advanced considerably over the last decade, the acute management of ICH remains limited to blood pressure control, coagulopathy reversal, and perhaps surgical evacuation of hemorrhage.^{3,4} These efforts have focused on reducing mortality and preventing additional morbidity due to secondary injury.

Clinical presentations for patients with stroke are varied based on the hemisphere affected, where hemispheric dominance dictates cognitive processing, attention, language, and emotional responses. Consequently, when confronted with ICH, the specific hemisphere affected can profoundly influence not only the clinical presentation but also management decisions and prognostic outcomes.

Previous research involving patients with ischemic stroke provide evidence that those with right hemispheric injury present later to the hospital and may not be treated as frequently with thrombolytic therapy.⁵⁻⁸ Some practitioners hesitate to offer decompressive craniectomy in patients with left dominant-hemispheric ischemic infarcts despite literature demonstrating that long-term functional outcomes and health-related quality of life (HRQoL) are equivalent if not better for patients with left hemispheric injuries.⁹⁻¹²

Amidst an increasing enthusiasm for surgical management of ICH,^{3,4,13} it is important to understand if hemispheric lateralization affects surgical decision making, as it seems to do for ischemic stroke. We aimed to evaluate the association of hemispheric lateralization of ICH on likelihood of receiving aggressive surgical intervention as well as its association with functional outcomes.

Methods

Study Design and Participants

We conducted a post-hoc analysis using data from two prospective studies: the Antihypertensive Treatment of Acute Cerebral Hemorrhage II (ATACH-2) trial and the Ethnic/Racial Variations of

Intracerebral Hemorrhage (ERICH) study. ATACH-2 was a large, multicenter, international, open-label, blinded-endpoint randomized controlled trial that enrolled 1000 patients with ICH from 2011 to 2016 and randomized them to either intensive blood pressure control (110-139 mm Hg) or standard blood pressure control (140-179 mm Hg). ERICH was a large, multicenter, prospective observational study that enrolled patients in the United States admitted with ICH from 2010 to 2017, with particular emphasis on recruitment of a racially and ethnically diverse cohort. The protocols and results of both studies have been previously published elsewhere.^{14,15}

We included all patients enrolled in these two studies with supratentorial location of ICH. We excluded patients with cerebellar or brainstem ICH as well as any patient with primary intraventricular hemorrhage (IVH).

The study was approved by the institutional review board of the Wake Forest School of Medicine (IRB00098198).

Exposures and Outcomes

Our exposure was hemispheric laterality (left versus right). Our primary outcome was neurosurgical intervention as a binary variable defined as receiving hematoma evacuation, decompressive craniectomy, or other craniotomy.

To explore differences in other aspects of aggressive care, we defined several secondary outcomes: receipt of external ventricular drain (EVD), receipt of tracheostomy, days of mechanical ventilation, receipt of do-not-resuscitate (DNR) order at any point in the hospitalization, withdrawal of life-sustaining therapy (WLST) at any point in the hospitalization, and early WLST (defined as < 72 from symptom onset).

Our primary functional outcome was modified Rankin score (mRS) at 90 days analyzed via ordinal regression. We also examined quality of life as assessed by the European Quality of Life-5 dimensions (EQ-5D) questionnaire.

Statistical Analysis

To estimate the association of hemispheric laterality on the outcomes of interest, we fitted hierarchical Bayesian regression models adjusted for the specified covariates below. We also included the study from which the patient was enrolled as a random intercept. We employed Bayesian statistical methods for this analysis due to several advantages over traditional frequentist approaches. Bayesian methods allow for the incorporation of prior knowledge or beliefs into the analysis, which can be particularly useful in observational studies where prior information about relationships between variables may exist. This is achieved through the use of prior distributions, which are combined with the observed data to produce posterior distributions representing updated beliefs about the parameters of interest. Furthermore, Bayesian methods provide a natural framework for quantifying uncertainty through credible intervals, which offer a more intuitive interpretation than frequentist confidence intervals. Finally, Bayesian modeling facilitates the estimation of posterior probabilities, allowing for direct probabilistic statements about hypotheses, such as the probability that an odds ratio is greater than 1, which is often of greater clinical relevance than a p-value.^{16,17}

This report follows suggestions from the Bayesian Analysis Reporting Guidelines.¹⁸ For binary endpoints, hierarchical Bayesian logistic regression models were used. For ordinal endpoints, hierarchical Bayesian ordinal logistic regression models were used. All analyses were performed in R version 4.4.2 (R Foundation for Statistical Computing, Vienna, Austria) using the ‘brms’ and ‘tidybayes’ packages.

Covariate Selection

We constructed directed acyclic graphs (DAGs) to model our causal assumptions and to appropriately select variables to minimize confounding and selection bias.^{19–21} Based on our specifications, we included the following covariates in our models: age, admission Glasgow coma score (GCS), ICH location (categorized as lobar, thalamic, or basal ganglia), baseline ICH volume (measured in milliliters), and presence of intraventricular hemorrhage (IVH). The DAGs used in the analyses are available in the supplementary material.

Prior Distribution Specification

We applied four sets of prior distributions for all models to assess the robustness of our findings. The first set was a neutral prior, centered on a log-odds ratio of 0, which corresponds to an odds ratio of 1 and assumes no difference in the outcome of interest between left and right hemispheric patients. The second and third sets were left hemisphere and right hemisphere priors, respectively, which were weakly informative priors that assumed a higher probability of the outcome occurring in the specified hemisphere. The final set was a flat prior, which assumes that all possible values for the model parameters are equally likely. This non-informative prior allows the data to dominate the posterior distribution and most closely approximates the assumptions of a frequentist analysis. More details on the rationale of prior specification and the specific parameters used for each prior distribution are included in the supplementary appendix.

Computation and Description of the Posterior Distribution

We report the following metrics for the association of hemispheric laterality on the primary and secondary outcomes: 1) median odds ratio (OR) of the posterior distribution with 95% credible interval; 2) probability of any difference ($P[OR] > 1$); 3) probability of a substantial difference ($P[OR] > 1.2$) as defined by the study authors; 4) percent of the posterior distribution outside of the region of practical equivalence as defined by the study authors (ROPE; $0.95 > P[OR] > 1.05$). A description of the rationale for these thresholds is available in the supplementary material.

Missing Data Treatment and Sensitivity Analyses

For our primary analysis, we used a complete case approach. To assess the robustness of our findings, we conducted a sensitivity analysis using multiple imputation with chained equations (MICE) to address missing data for all covariates.

In addition to the missing data analysis, we analyzed our models using the four sets of prior distributions described above, which allowed us to examine the influence of prior choices on the posterior

distributions. Further details on the methods and results of these sensitivity analyses are presented in the Supplementary Appendix.

Data Availability Statement

The ATACH-2 (PI: Adnan I. Qureshi, MD, NCT01176565) and ERICH (PI: Daniel Woo, MD, MS, NCT01202864) data and analyses presented in this manuscript are based on the research files downloaded from the Archived Clinical Research web site of the National Institute of Neurologic Disease and Stroke (NINDS). The datasets can be obtained after approval of a written proposal as outlined on the NINDS website. As outlined in the NINDS Data Request Agreement, the analysis code is publicly available at <https://github.com/nikhilpatelmd/ich-laterality>.

Results

Baseline Characteristics

A total of 3477 patients were included in the analysis, of which 1720 (49%) had left hemispheric ICH (Table 1). They had higher National Institutes of Health Stroke Scale (NIHSS) scores, lower GCS scores, and lower rates of hypertension as compared to those with right hemispheric ICH. There were 2487 (72%) patients in the cohort from ERICH and 990 (28%) from ATACH-2.

Hemispheric laterality and neurosurgical intervention

Associations between hemispheric laterality and primary and secondary outcomes are shown in Table 2. There were 86 (5.0%) with left hemispheric ICH who underwent neurosurgical intervention as compared to 144 (8.2%) with right hemispheric ICH (aOR 1.56, 95% credible interval 1.07 - 2.26). The posterior probability that the aOR was greater than 1 was 99%. The posterior probability of the aOR being greater than 1.2, indicating a substantial difference, was 92%. A very small portion of the posterior probability was inside of the ROPE (2%).

The results using different pre-defined priors were consistent with the findings in the main model (Figure 1). Sensitivity analyses yielded similar results (Supplemental Appendix).

Hemispheric laterality and secondary outcomes

A total of 236 (13.7%) with left hemispheric ICH had an external ventricular drain (EVD) placed as compared to 272 (15.5%) with right hemispheric ICH (aOR 1.34, 95% credible interval 1.01 - 1.76). The posterior probability that the aOR was greater than 1 was 98%. The posterior probability that the aOR was greater than 1.2 was 78%. About 2% of the posterior distribution was inside of the ROPE.

Other secondary outcomes showed smaller and less certain associations with hemispheric laterality.

Subgroup Analyses

The results of the pre-specified subgroup analyses are shown in Table 3.

ICH Location

While the probability of neurosurgical intervention in patients with right hemisphere ICH was consistent when considering different locations, the difference was greatest among patients with lobar ICH. In patients with lobar ICH, a total of 39 (7.7%) patients with left hemispheric ICH underwent neurosurgical intervention as compared to 76 (14.9%) patients with right hemispheric ICH (aOR 2.00, 95% credible interval 1.30 - 2.99). The posterior probability that the aOR was greater than 1 was 100%. The posterior probability that the aOR was greater than 1.2 was 99%. The entire posterior distribution was outside of the ROPE.

Study

The association of hemispheric laterality was bigger in the ERICH cohort as compared to the ATACH-2 cohort. Of the patients enrolled in ERICH, 73 (5.8%) patients with left hemispheric ICH underwent neurosurgical intervention as compared to 114 (9.2%) patients with right hemispheric ICH (aOR 1.86, 95% credible interval 1.27 - 2.64). The posterior probability that the aOR was greater than 1 was 100%.

The posterior probability that the aOR was greater than 1.2 was 99%. The entire posterior distribution was outside of the ROPE.

Hemispheric laterality and functional outcomes

The associations between mRS at 90 days and functional outcomes are described in Table 4. The aOR for a worse mRS score at 90 days associated with right hemispheric ICH compared to left hemispheric ICH was 1.18 (95% credible interval 1.01 - 1.36). The posterior probability that the aOR was > 1 was 98%. The posterior probability that the aOR was > 1.2 was 39%. About 6% of the posterior distribution was inside of the ROPE.

When examining HRQOL, the largest differences between groups was in the Mobility and Self-Care domains. The aOR for a worse EuroQOL - Mobility score associated with right hemispheric ICH compared to left hemispheric ICH was 1.43 (95% credible interval 1.2 - 1.71) with similar results for the Self-Care domain. The posterior probability that the aOR was > 1 was 100% for both domains, with the posterior probability that the aOR was > 1.2 being 98%.

Results from using different priors and sensitivity analyses are presented in the Supplementary Appendix and did not significantly alter the results.

Discussion

In this pooled analysis of 3477 patients with ICH, we found evidence that those with right hemispheric ICH were more likely to receive neurosurgical intervention than those with left hemispheric ICH. Specifically, patients with right hemispheric ICH had 1.56 times the odds of receiving a neurosurgical intervention and 1.34 times the odds of receiving an EVD. This was consistent when considering various ICH locations and within each individual study. Despite receiving less aggressive surgical care, patients with left hemispheric ICH had better 90-day functional outcomes, a finding robust to various definitions including the mRS and HRQoL domains.

This finding is particularly important in light of recent evidence suggesting that surgical interventions, such as hematoma evacuation and decompressive craniectomy, may improve outcomes in some ICH patients.^{3,13} If there is a bias against offering these interventions to patients with left hemispheric ICH, it could lead to worse outcomes for this group. This highlights the need to address potential biases in treatment decisions and ensure equitable access to life-saving and disability-improving interventions. While treatment bias based on hemispheric laterality is not well-characterized in ICH, there are several studies that have found that clinicians, patients, and families place particular importance in language deficits and thus are less optimistic about dominant-hemispheric injuries.^{22–24} This attitude may not be warranted if patients with left hemispheric injury report equivalent or improved quality of life outcomes compared to those with right hemispheric injury.

Evidence is limited regarding the association of hemispheric lateralization and functional outcomes in patients with ICH. Our findings corroborate those from a pooled analysis of the INTERACT1 and INTERACT2 trials, which showed a higher risk of mortality in patients with right hemispheric ICH.¹² Most other studies examining the association of hemispheric laterality and functional outcomes have focused on patients with acute ischemic stroke with mixed results. The two largest studies were pooled analyses from prospective ischemic stroke trials and neither found a clear association between hemispheric laterality and outcomes.^{25,26} This may be particularly challenging to study in ischemic stroke due to how the NIHSS is used to guide treatment decisions and is also biased toward higher scores in patients with left hemispheric injury.²⁷

Our study has some limitations. The absolute number of patients undergoing surgery was relatively low, making it unclear how these findings might translate to studies with higher intervention rates. We derived our cohort from two prospective studies where practice patterns may have been affected by the Hawthorne effect. Our cohort is also limited to mostly patients living in the United States, and there is likely to be a range of different socioeconomic and cultural frameworks that factor into decisions on treatment, prognosis, and quality of life throughout the world. With the recent publication of several large

RCTs demonstrating the benefit of hematoma evacuation and decompressive craniectomy in ICH,^{3,13} it will be imperative to confirm our findings with additional cohorts. Future research should focus on prospectively surveying clinicians, patients, and families with a variety of cultural and socioeconomic backgrounds as well as in a range of practice settings to further investigate the potential spectrum of biases in treatment decisions and their impact on outcomes.

Conclusions

We found that patients with left hemispheric ICH were less likely to receive aggressive care in the form of neurosurgical intervention and external ventricular drain placement as compared to those with right hemispheric ICH. Despite this, patients with left hemispheric ICH were more likely to have better functional outcomes at 90 days. These findings highlight the potential influence of hemispheric laterality on treatment decisions and outcomes in ICH and suggest that further research is needed to understand and address potential biases in treatment allocation.

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Conflict of Interest Disclosures: Dr. Magun discloses being a shareholder in Inovio Pharmaceuticals and StageZero Life Sciences.

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Table 1 – Baseline Characteristics Stratified by Hemispheric Laterality

Characteristic	Overall N = 3,477 ¹	Left N = 1,720 ¹	Right N = 1,757 ¹	p-value ²
Age	61 (52, 73)	61 (52, 72)	61 (52, 73)	0.91
Sex				0.12
Male	2,081 (60%)	1,007 (59%)	1,074 (61%)	
Female	1,396 (40%)	713 (41%)	683 (39%)	
Race				0.59
White	1,875 (54%)	935 (54%)	940 (54%)	
Black or African-American	1,022 (29%)	510 (30%)	512 (29%)	
Asian	557 (16%)	263 (15%)	294 (17%)	
Other/Not Reported	16 (0.5%)	7 (0.4%)	9 (0.5%)	
American Indian or Alaskan Native	7 (0.2%)	5 (0.3%)	2 (0.1%)	
Ethnicity				0.42
Not Hispanic or Latino	2,296 (66%)	1,141 (66%)	1,155 (66%)	
Unknown/Not Reported	1,052 (30%)	509 (30%)	543 (31%)	
Hispanic or Latino	129 (3.7%)	70 (4.1%)	59 (3.4%)	
LNW to ED Presentation (mins)	78 (60, 180)	74 (60, 180)	83 (59, 180)	0.20
Baseline SBP (mm Hg)	190 (165, 215)	190 (165, 216)	190 (165, 214)	0.34
Baseline NIHSS	11 (6, 16)	11 (6, 17)	11 (7, 14)	0.014
Baseline GCS	15 (12, 15)	14 (11, 15)	15 (13, 15)	<0.001
Hypertension	784 (23%)	366 (21%)	418 (24%)	0.076
Type II Diabetes	705 (20%)	348 (20%)	357 (20%)	0.95
Previous Stroke	583 (17%)	293 (17%)	290 (17%)	0.68
Tobacco Use	1,361 (39%)	663 (39%)	698 (40%)	0.48
ICH Location				0.002
Basal Ganglia	1,304 (38%)	688 (40%)	616 (35%)	
Thalamus	1,157 (33%)	527 (31%)	630 (36%)	
Lobar	1,016 (29%)	505 (29%)	511 (29%)	
ICH Volume (mL)	11 (5, 26)	11 (5, 24)	12 (5, 27)	0.060

IVH	1,304 (38%)	624 (36%)	680 (39%)	0.14
Study				0.18
ERICH	2,487 (72%)	1,248 (73%)	1,239 (71%)	
ATACH-2	990 (28%)	472 (27%)	518 (29%)	
¹ Median (Q1, Q3); n (%)				
² Wilcoxon rank sum test; Pearson's Chi-squared test; Fisher's exact test				

Table 2 – Association of Hemispheric Laterality with Primary and Secondary Outcomes

Outcome	Left Hemisphere	Right Hemisphere	aOR (95% CI) ¹	Probability of any difference (aOR > 1)	Probability of a substantial difference (aOR > 1.2)	ROPE ²
Neurosurgical Intervention	86 (5.0%)	144 (8.2%)	1.56 (1.07 - 2.26)	0.99	0.92	0.02
External Ventricular Drain	236 (13.7%)	272 (15.5%)	1.34 (1.01 - 1.76)	0.98	0.78	0.04
Days of Mechanical Ventilation	6 (3 - 12) ³	6 (3 - 9) ³	0.94 (0.49 - 1.78)	0.43	0.23	0.12
Tracheostomy	19 (1.1%)	19 (1.1%)	1.05 (0.63 - 1.76)	0.58	0.31	0.15
Withdrawal of Life-Sustaining Therapy	152 (8.8%)	143 (8.1%)	1.01 (0.72 - 1.4)	0.52	0.15	0.23
Early Withdrawal of Life-Sustaining Therapy	45 (2.6%)	39 (2.2%)	0.91 (0.58 - 1.41)	0.33	0.11	0.16
DNR Order	195 (11.3%)	202 (11.5%)	1.06 (0.77 - 1.45)	0.63	0.21	0.23

¹aOR = adjusted odds ratio, CI = 95% credible interval; adjusted for age, admission GCS, ICH location, ICH volume, IVH, and study (as random intercept);
Reference Category: Left Hemisphere Laterality

²ROPE = region of practical equivalence, defined as 0.95 > aOR > 1.05

³median (IQR)

Table 3 – Subgroup Analysis of Neurosurgical Intervention by ICH Location and Study

Subgroup	Left Hemisphere	Right Hemisphere	aOR (95% CI) ¹	Probability of any difference (aOR > 1)	Probability of a substantial difference (aOR > 1.2)	ROPE ²
ICH Location						
Basal Ganglia	43 (6.2%)	51 (8.3%)	1.39 (0.86 - 2.12)	0.91	0.70	0.07
Lobar	39 (7.7%)	76 (14.9%)	2.00 (1.30 - 2.99)	1.00	0.99	0.00
Thalamus	4 (0.8%)	17 (2.7%)	1.79 (0.83 - 3.42)	0.92	0.82	0.04
Study						
ATACH-2	13 (2.8%)	30 (5.8%)	2.83 (1.31 - 5.45)	1.00	0.99	0.00
ERICH	73 (5.8%)	114 (9.2%)	1.86 (1.27 - 2.64)	1.00	0.99	0.00

¹aOR = adjusted odds ratio, CI = 95% credible interval; adjusted for age, admission GCS, ICH location, ICH volume, IVH, and study (as random intercept);
Reference Category: Left Hemisphere Laterality

²ROPE = region of practical equivalence, defined as 0.95 > aOR > 1.05

Table 4 – Association of Hemispheric Laterality with 90 Day Functional Outcomes

Outcome	aOR (95% CI)¹	Probability of any difference (aOR > 1)	Probability of a substantial difference (aOR > 1.2)	ROPE²
Modified Rankin Score	1.18 (1.01 - 1.36)	0.98	0.39	0.06
EuroQOL - Mobility	1.43 (1.2 - 1.71)	1.00	0.98	0.00
EuroQOL - Self-Care	1.43 (1.2 - 1.72)	1.00	0.98	0.00
EuroQOL - Usual Activities	1.17 (0.98 - 1.39)	0.96	0.37	0.11
EuroQOL - Pain/Discomfort	1.24 (1.05 - 1.48)	0.99	0.66	0.03
EuroQOL - Anxiety/Depression	1.16 (0.97 - 1.38)	0.95	0.35	0.12
Euro VAS	0.95 (0.88 - 1.03)	0.87 ³	0.00 ⁴	0.52
¹ aOR = adjusted odds ratio, CI = 95% credible interval; adjusted for age, admission GCS, ICH location, ICH volume, IVH, and study (as random intercept); Reference Category: Left Hemisphere Laterality				
² ROPE = region of practical equivalence, defined as 0.95 > aOR > 1.05				
³ Probability of aOR < 1				
⁴ Probability of aOR < 0.8				